

Zach James

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EDUCATION

Louisiana State University

BS in Mathematics, BS in Computer Science, Minor in Psychology

GPA: 3.69 CS / 3.45 Overall

Baton Rouge, LA

Graduation: May 2026

- Organizations/Awards: Honors College, Dean's List, President's Honor Roll, S&B Engineers Scholarship, LSU ICPC Competitive Programming team, Startup LSU
- Selected Coursework: Abstract Algebra II, Applied Optimization Theory, Numerical Linear Algebra, Operating Systems, Computer Networks, Malware Analysis, AI, Math Stats

WORK EXPERIENCE

LSU Department of Mathematics

Undergraduate Research Assistant

Baton Rouge, LA

May 2024 – March 2026

- NSF-Funded TDA Project: Developed Vietoris-Rips and alpha complex construction pipelines; benchmarked persistence computations across coordinate, Isomap, and UMAP embeddings
- Clinical ML Project: Co-authored biomedical ML research accepted by Frontiers in Applied Mathematics and Statistics; publication in process
- Trained supervised and semi-supervised models, including p-Laplacian graph regression, on biomarker data to predict ALM, BMD, and BFP using compute clusters

Brick Bridge Consulting

Software Developer

Louisville, KY

December 2022 – January 2024

- Designed and independently delivered HRM, billing, and financial systems for a medical staffing firm using JavaScript and Podio workflow automation
- Built internal analytics and financial tools for client businesses, including invoice generation, rate-card calculators, tax logic, and reporting dashboards
- Developed multi-interface CRUD application for a medical-legal startup, including PHP API integrations, HTML/CSS frontend, and leveled access control

LSU Division of Computer Science & Engineering

Undergraduate Teaching Assistant

Baton Rouge, LA

December 2022 – May 2024

- TA for 100+ student Java programming labs; grading, proctoring, one-on-one tutoring

PROJECTS

Thymos — Compositional C ML Runtime

- Designing and prototyping a zero-dependency C runtime for typed tensor graphs, with explicit memory ownership, context-driven allocation, and backend-dispatch structure for scalar CPU execution first
- Building flat SSA-style graph IR inspired by applied category theory, with typed values, op nodes, validation, execution planning, and rewrite/fusion-oriented design

Reverse Engineering & Malware Writeups

- Published technical writeups at resume.agapebloom.com/writeups on SQL Slammer shellcode, DOS/boot-sector malware, and x86_64 bot analysis, covering IDA/Ghidra workflows, import resolution, payload logic, and anti-debug behavior
 - Solved all 0x40 xorpd x86 assembly puzzles ("little black book"): github.com/zach15james/xorpd
- Completed Malware Analysis & RE coursework covering x86 assembly, binary analysis, Ghidra/IDA workflows, and malware walkthroughs

SKILLS

- **Languages:** C, C++, Python, JavaScript, Java, SQL
- **Systems/Tools:** Linux, Git, CMake, CUDA, Ghidra, IDA Pro, pwn.college
- **Technical:** statistical learning, optimization, data pipelines, RE, arena allocation, multithreading
- **Interests:** low-level systems, embedded ML, quantitative research, reverse engineering